Customer Repeat Purchase Analysis

Level 2 • Day 28 | Data Analytics Track

Objective: Measure repeat purchase behavior, identify repeat-customer segments, and understand customer loyalty.

Metric	Result
Total customers	2,500
Repeat customers	1,299
One-time customers	1,201
Repeat customer rate	51.96%
First-time AOV	52.36
Repeat customer AOV	65.83

Methodology

A repeat customer is defined as a customer with at least two distinct orders during the analysis period. Customer-level metrics were created by grouping transactions by Customer_ID. Average Order Value (AOV) was calculated as total customer sales divided by the number of distinct orders.

Customer Segmentation

Segment	Customers	Total Sales	Avg Orders	Avg AOV	Cust. Share	Sales Share
Loyal (4+ orders)	140	46,899.94	4.30	77.38	5.60%	16.70%
One-time	1201	62,889.09	1.00	52.36	48.04%	22.40%
Repeat (2-3 orders)	1159	170,992.86	2.27	64.44	46.36%	60.90%

First-time vs Repeat AOV

Repeat customers have an average order value of 65.83, compared with 52.36 for one-time customers. This represents an AOV difference of 25.7%, indicating that retaining customers can increase the value of subsequent purchases.

Key Insights

1. 51.96% of customers placed at least two orders, giving a direct measure of customer retention/loyalty.

2. Repeat customers show a higher average order value (65.83) than one-time customers (52.36).

3. The One-time segment is the primary acquisition opportunity: targeted second-purchase campaigns can move customers into the repeat segment.

4. The Repeat (2-3 orders) segment is the key conversion pool for loyalty programs, personalized recommendations, bundles, and post-purchase offers.

5. Loyal (4+ orders) customers should receive retention-focused treatment because they have demonstrated strong purchase frequency.

Business Recommendations

• Trigger a second-purchase campaign 15–30 days after the first order.

• Use product recommendations based on the customer's previous category and basket.

• Offer loyalty points or tiered benefits to customers reaching 2–3 orders.

• Create win-back campaigns for customers whose last purchase is becoming inactive.

• Monitor repeat rate and repeat AOV monthly to evaluate retention initiatives.

Interview Questions — Suggested Answers

How do you define a repeat customer?

A customer who has completed at least two distinct orders during the analysis period.

Why track repeat rate?

Repeat rate measures customer loyalty and retention. It helps evaluate whether acquisition is creating customers who return and generate additional revenue.

Technical Deliverables

The accompanying project folder contains the transaction dataset, customer-level segment table, SQL queries, Python analysis script, category summary, README, and this PDF report.